

• STRATEGY GUIDE

Token-Efficient LLM Use

Spend tokens where they produce work, not where habit burns them.

7

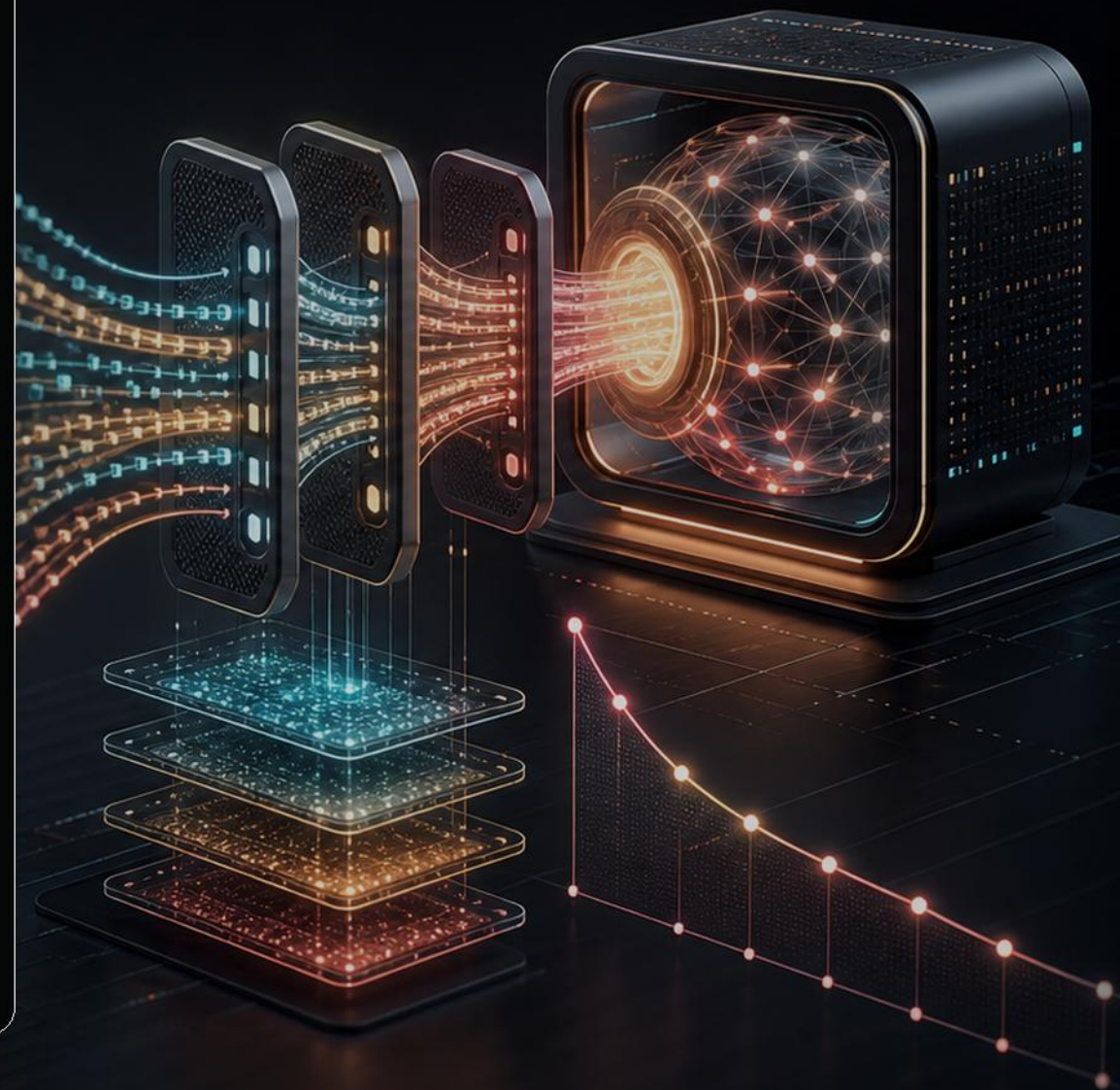
STACKING LEVERS

90%

CACHE READ
DISCOUNT

50-98%

ROUTING SAVINGS



• THE PROBLEM

Usage grows fast. Budgets do not.

The goal is not to slow adoption. The goal is to make the default path cheaper.

- Token volume rises as teams rely on AI more often.
- Hard caps create friction and kill useful adoption.
- Better defaults let usage grow while spend stays controlled.

67%

ONE-GENERATION
OPUS PRICE DROP

90%

CACHE READ
DISCOUNT

98%

ROUTING UPSIDE



• THE PLAYBOOK

Seven levers that stack

Each lever helps on its own. Together they compound.

- 1. Route to the cheapest adequate model.
- 2. Tune reasoning effort to task difficulty.
- 3. Cache stable prompt prefixes.
- 4. Compress noisy context before inference.
- 5. Keep agents away from context-rot cliffs.
- 6. Use code for deterministic routine work.
- 7. Measure useful output per token.



• LEVER 1

Model routing

Do not default every request to the most expensive model.

- Simple formatting and extraction can use cheap models.
- Standard work belongs on mid-tier models.
- Reserve frontier models for planning and hard reasoning.

Question: what is the cheapest model that is good enough?



• LEVER 2

Effort tuning

Reasoning depth is a budget knob.

- Low effort is enough for simple or bulk tasks.
- Medium is the best default for most work.
- Max effort should be deliberate, not habitual.

Default to medium. Escalate only when the task earns it.



• LEVER 3

Prompt caching

Pay once for stable context, then reuse it.

- Put static content first and dynamic content last.
- Keep prefixes byte-for-byte identical across requests.
- Pre-warm long-running agent sessions when possible.

Cache reads can be dramatically cheaper than fresh input.



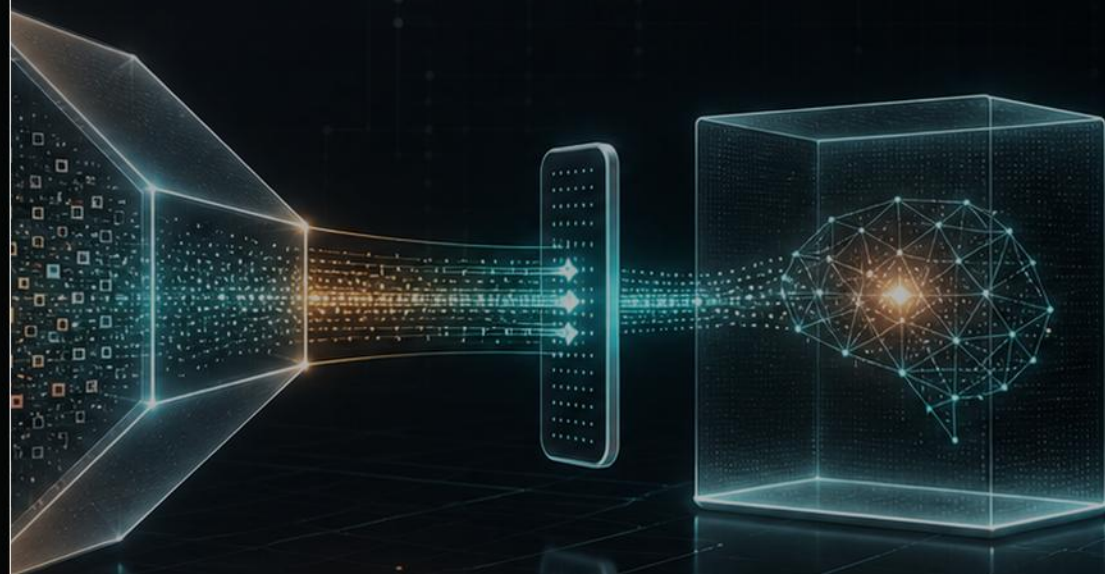
• LEVER 4

Context compression

Send signal, not sludge.

- Strip irrelevant logs, repeated files, and stale discussion.
- Keep exact names, dates, and numbers when they matter.
- Use compression to create headroom before the model sees context.

Best use: reduce noise while preserving the task signal.



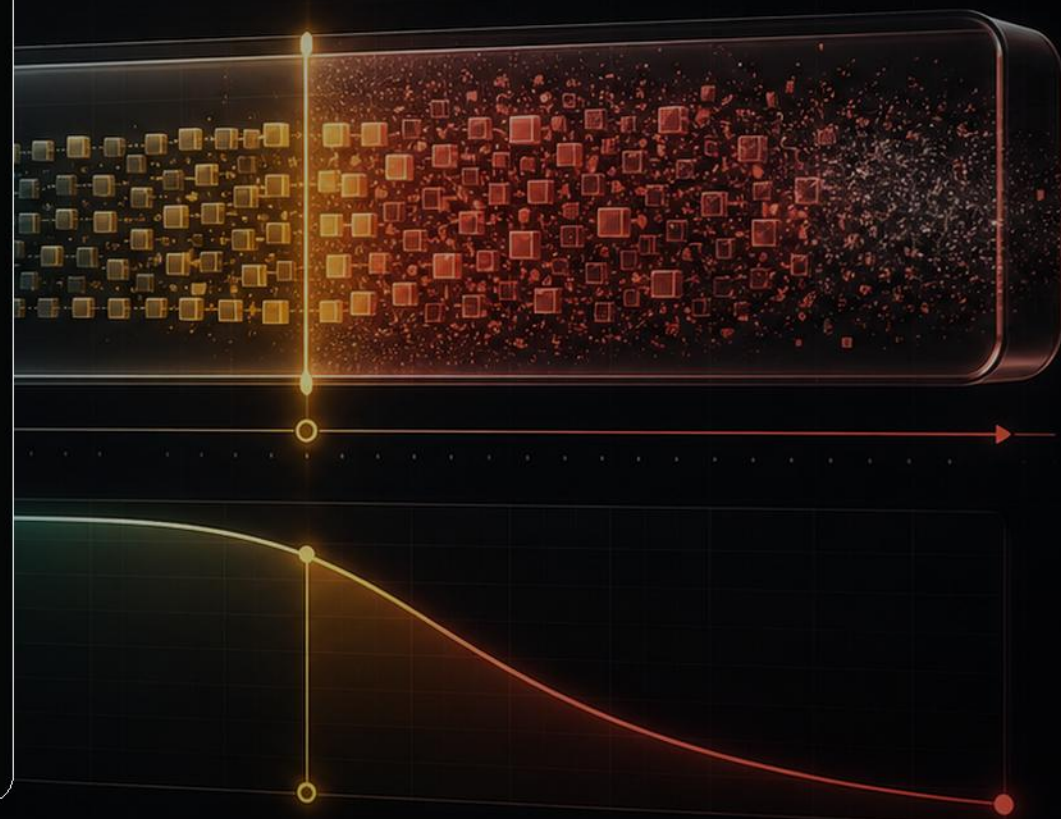
• LEVER 5

Context rot

A bigger window is not the same as better thinking.

- Quality often drops before the context window is full.
- Split or compact work before the session gets crowded.
- Size agent tasks so they finish before the quality cliff.

Rule of thumb: watch the 60% fill zone.



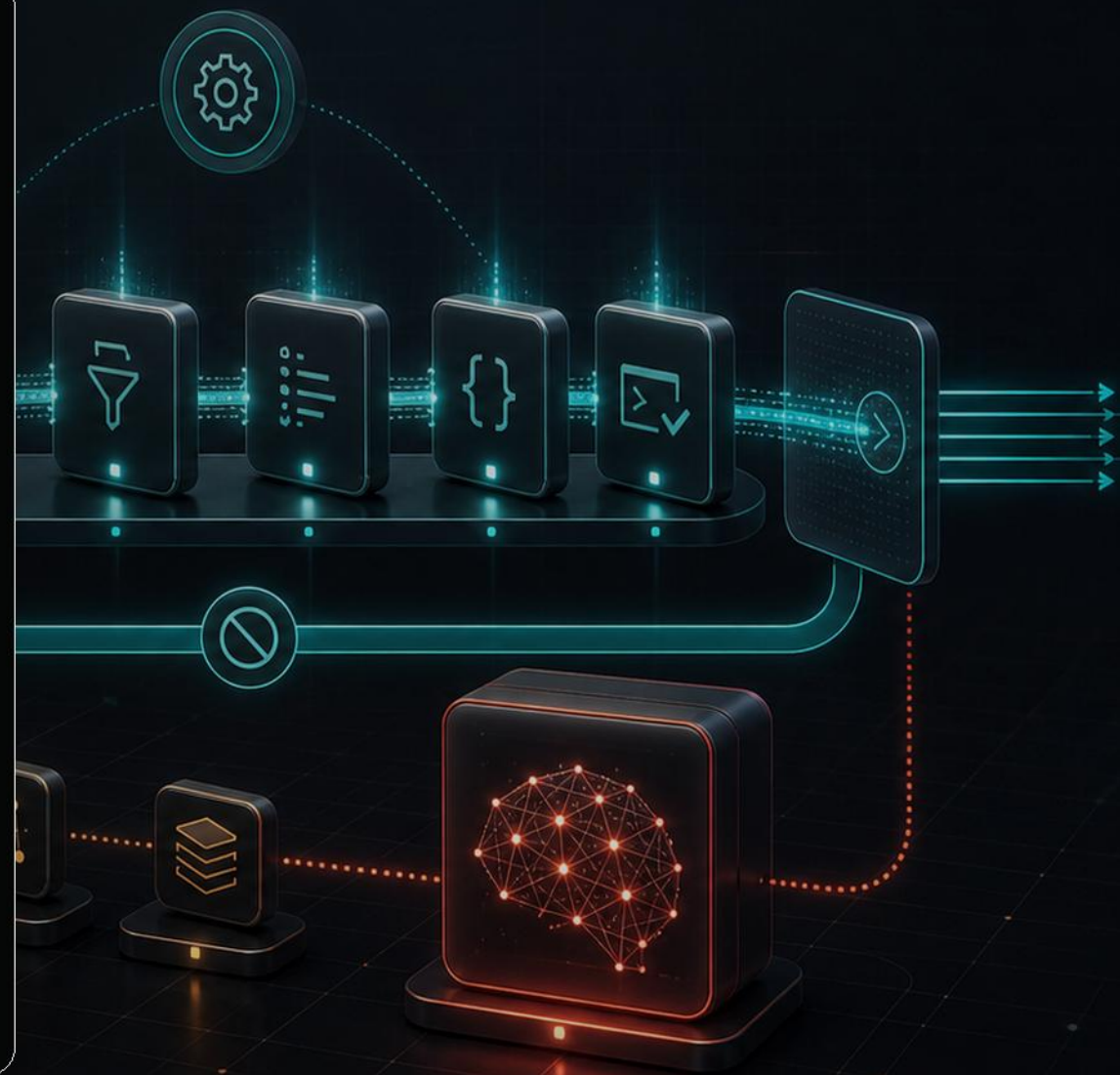
• LEVER 6

Code over inference

The cheapest token is the one you never send.

- Use scripts for formatting, parsing, validation, and merging.
- Keep AI for judgment, ambiguity, and planning.
- Turn repeated prompt work into deterministic tools.

Routine work should become code, not recurring model calls.



• LEVER 7

Tokenmaxxing

Spend more only when more tokens buy more output.

- Measure useful output per token, not token volume alone.
- Spend on retrieval, review, tests, and verification.
- Cut tokens that do not improve the result.

Less waste, not less ambition.



• CASE STUDY

Platform discipline in practice

Let usage grow while the platform controls waste.

- Gateway routes by price, cache state, and task need.
- Cheap defaults handle execution; frontier models handle planning.
- Caching, compression, and code reduce waste.
- Teams measure useful output, not token volume alone.

The pattern: freedom for engineers, discipline in the platform.



• TAKEAWAY

Fewer tokens wasted. Not fewer tokens used.

Route smarter. Cache aggressively. Compress ruthlessly. Code what you can. Spend tokens that earn their keep.

Route

BEFORE YOU RUN

Cache

BEFORE YOU
REPEAT

Code

BEFORE YOU INFER

